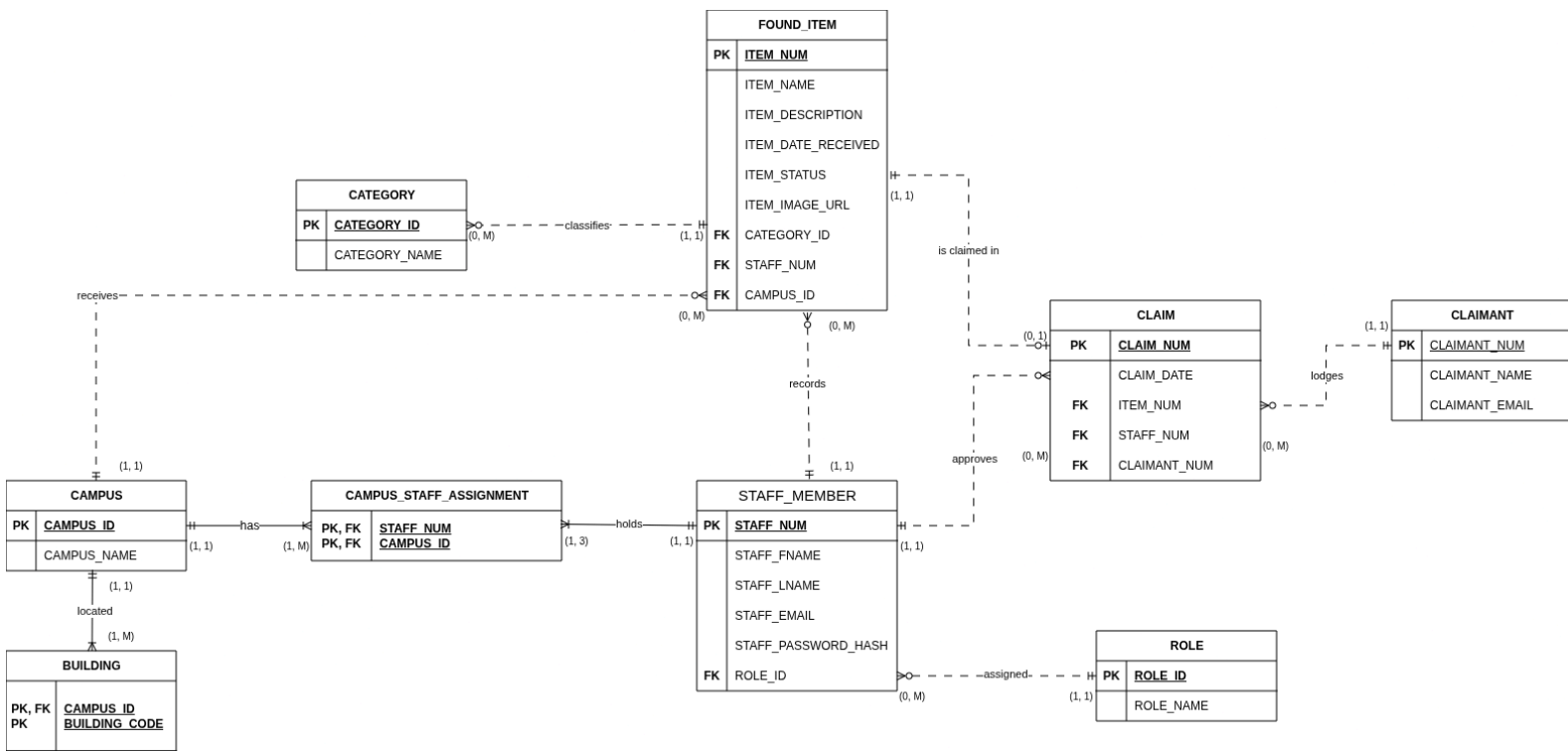


1. Conceptual Design

Business rules derived from Operations

1. A Campus contains one or many Buildings. A Building is located at exactly one Campus.
2. A Campus employs one or many Protection Services staff members (STAFF_MEMBER). A Protection Services staff member (STAFF_MEMBER) is assigned to one or a maximum of 3 Campuses.
3. A Protection Services staff member (STAFF_MEMBER) is assigned exactly one Role. A Role may be assigned to zero or many Protection Services staff members (STAFF_MEMBER)
4. A Found Item is classified under exactly one Category. A Category classifies zero or many Found Items.
5. A Found Item is received and stored at exactly one Campus. A Campus receives 0 or many Found Items.
6. A Protection Services staff member (STAFF_MEMBER) records zero or many Found Items. A Found Item is logged by exactly one Protection Services staff member (STAFF_MEMBER).
7. A Claim is lodged by exactly one Claimant. A Claimant can lodge zero or many Claims over time.
8. A Found Item may be claimed in zero or one Claim. A Claim is made for exactly one Found Item.
9. A Claim is verified and approved by exactly one Protection Services staff member (STAFF_MEMBER). A Protection Services staff member (STAFF_MEMBER) can verify zero or many Claims.

Entity Relationship Diagram



It is difficult to fit the ERD image clearly, please view the quality image here:
<https://drive.proton.me/urls/D1HJFJGBGG#bCDoYtWNWR96>

Weak Entities: BUILDING, CAMPUS_STAFF_ASSIGNMENT

Composite Entities: CAMPUS_STAFF_ASSIGNMENT

Composite Keys: In BUILDING and CAMPUS_STAFF_ASSIGNMENT

Weak relationships: Denoted with solid lines

Strong relationships: Denoted with dashed lines

Optional relationships: Denoted with (0, M), (0, 1) and (0, M) cardinalities

Mandatory relationships: Denoted with (1, 1) and (1, M) cardinalities

2. Logical Data Model

CAMPUS (CAMPUS_ID, CAMPUS_NAME)

PRIMARY KEY: CAMPUS_ID

BUILDING (CAMPUS_ID, BUILDING_CODE)

PRIMARY KEY: CAMPUS_ID, BUILDING_CODE

FOREIGN KEY: CAMPUS_ID REFERENCES CAMPUS

ROLE (ROLE_ID, ROLE_NAME)

PRIMARY KEY: ROLE_ID

STAFF_MEMBER (STAFF_NUM, STAFF_FNAME, STAFF_LNAME, STAFF_EMAIL, STAFF_PASSWORD_HASH, *ROLE_ID*)

PRIMARY KEY: STAFF_NUM

FOREIGN KEY: *ROLE_ID* REFERENCES ROLE

CAMPUS_STAFF_ASSIGNMENT (STAFF_NUM, CAMPUS_ID)

PRIMARY KEY: STAFF_NUM, CAMPUS_ID

FOREIGN KEYS: STAFF_NUM REFERENCES STAFF_MEMBER

CAMPUS_ID REFERENCES CAMPUS

CATEGORY (CATEGORY_ID, CATEGORY_NAME)

PRIMARY KEY: CATEGORY_ID

FOUND_ITEM (ITEM_NUM, ITEM_NAME, ITEM_DESCRIPTION, ITEM_DATE_RECEIVED, ITEM_STATUS, ITEM_IMAGE_URL, *CATEGORY_ID*, *STAFF_NUM*, *CAMPUS_ID*)

PRIMARY KEY: ITEM_NUM

FOREIGN KEYS: *CATEGORY_ID* REFERENCES CATEGORY

STAFF_NUM REFERENCES STAFF_MEMBER

CAMPUS_ID REFERENCES CAMPUS

CLAIMANT (**CLAIMANT_NUM**, CLAIMANT_NAME, CLAIMANT_EMAIL)

PRIMARY KEY: CLAIMANT_NUM

CLAIM (**CLAIM_NUM**, *ITEM_NUM*, CLAIM_DATE, *STAFF_NUM*, *CLAIMANT_NUM*)

PRIMARY KEY: CLAIM_NUM

FOREIGN KEYS: ITEM_NUM REFERENCES FOUND_ITEM

STAFF_NUM REFERENCES STAFF_MEMBER

CLAIMANT_NUM REFERENCES CLAIMANT